Real-Time Detection of Actual and Early Gait Events During Level-Ground and Ramp Walking

Saikat Sahoo, Mahesh Saboo, Dilip Kumar Pratihar, *Senior Member, IEEE,* Sudipta Mukhopadhyay, *Senior Member, IEEE,*

***Abstract*—A gait cycle is divided into stance and swing phases. The stance phase is further divided into several inner-stance phases. Detection of the inner stance phases helps to provide fine control to the powered prosthetic and assistive devices (PPADs). In recent studies on gait event detection, the inertial sensor-based approaches were found to be more reliable than that uses force sensors. However, inertial sensor-based algorithms exhibit some time delay. This time delay can affect the stability of the PPADs. Hence, several methods were proposed to reduce the same. However, it is impossible to completely eliminate the effect of delay using the existing methods. In this study, we proposed a novel early detection strategy to tackle the issue discussed above. The strategy involved the detection of both actual as well as early gait events. Thus, a total of nine gait events were detected in real-time using inertial sensors and a set of rules. The proposed method was tested for ten control subjects during both level ground as well as ramp walking, and the detection accuracy was found to be equal to 100%. Moreover, the time delay for the detection of actual gait events using the proposed method was found to be similar to some state-of-the-art methods and better than a few. The average early detection time was found to be equal to 123 ms that can be utilized to improve the response of PPADs.**

***Index Terms*—Finite state control; gait events; inertial measurement unit; lower limb prosthesis; real-time.**

I. Introduction

THE gait cycle was divided into several discrete phases, namely load acceptance, high impedance, high torque, low impedance and high velocity for controlling the kinetics of powered prosthetic and assistive devices (PPADs). This approach is known as the state transition model, and it is popular for controlling the lower limb PPADs [1], [2]. The low-level motor controller for PPADs does the job of switching between the two of the mentioned phases. However, in real-time, this can be achieved by a proper synchronization between the high and low-level controllers. The high-level controller is nothing but a gait phase detection system. Hence, the detection of gait phases is crucial and it is the most fundamental part of controlling the PPADs. Initially, only two gait events, such as heel strike (HS) and toe-off (TO) were utilized for the same. The HS and TO indicate the beginning of the stance and swing phases, respectively. However, the detection of these two phases is not sufficient to gain the better stability [3], [4]. Hence, the inner-stance phases i.e., the phases between the HS and TO events are essential to be detected for developing the state transition model. Foot flat (FF) and heel off (HO) events are popular for detecting the inner stance phases [3].

The commonly used sensor for gait phase detection is the force-sensitive resistors (FSR) [5]. Though a FSR gives an accurate measurement of the gait phase, it has issues like limited reliability, poor durability, and its sensitivity reduces after a few loading cycles [6], [7]. Moreover, it does not provide information about the inner-swing phases and has poor aesthetic property. These issues make it inefficient to be used for PPADs. For the last decade, the inertial measurement units (IMUs) had been used to get over the mentioned limitations, and it was found to be suitable for the purpose of prosthesis control. Therefore, some studies had been carried out for the detection of gait events using either accelerometer or gyroscope or a combination of them. Several algorithms like machine learning [8] and rule-based [9] approaches were explored for this purpose. The rule-based approaches were seen to be computationally faster, easier to implement and more suitable for real-time applications compared to machine learning approaches. The measurement of gait events using IMU exhibits some time delay, compared to that utilizing the FSR. The delay in detection along with controlling settling time may result into a sluggish response, which can cause instability while walking using the PPADs.

As discussed earlier, the maximum studies on gait event detection were focused on the detection of HS and TO [10]–[15]. A few recent studies explored the detection of the intermediated events between HS and TO using IMU. For example, Mannini et al. [16] proposed an online machine learning approach for the detection of gait events like foot strike, foot flat, HO and TO. The performance was tested for five able-bodied subjects walking at different speeds. However, the delay reported for HO was found to be high (i.e., $>$ 100 ms for 25% data). Similarly, the methods proposed in [17] and [18] reported a small time delay for the detection of HS, TS, HO and TO events for control subjects. However, these methods were tested offline, and hence, it is difficult to implement them in real-time. Lambrecht et al. [19] detected the gait events, such as initial contact, FF, HO and TO using the kinematic data along with a biomechanical model. The three threshold-based algorithms proposed by them showed either large variability or high mean delay for the detection of HO, whereas other events were detected with the smaller delays. Moreover, the results were evaluated offline using joint kinematics data. The method proposed by Muller et al. [20] dealt with the detection of initial contact, complete contact, HO and TO using one foot-mounted IMU. Though it was a real-time approach, the time delay reported in this study was found to be on the higher side. Maqbool et al. [21] proposed a real-time approach for the detection of initial contact, foot flat start, HO and TO events using one IMU attached to the shank

and utilizing a set of rules. The method was tested for level ground and ramp walking. Though the reported time delay was found to be small, the detection accuracy was reported less than 100% for HO. Han et al. [9] considered the detection of gait events, such as HS, toe strike and TO for detecting normal and abnormal gaits, while walking on the level ground. However, they reported a low detection accuracy. Another study reported in [22], used Bayesian recognition technique for the detection of eight gait phases. However, the information regarding time delay was not mentioned. The aim of the above mentioned studies was to increase the detection accuracy and minimize the delay. A few of them achieved a detection accuracy of nearly 100%. However, a complete elimination of time delay is not possible using the above mentioned approaches. In real-time, this might affect the performance of PPADs.

One way of solving the issue related to time delay could be an early detection technique. In this study, a method that deals with actual and early detection of gait events was proposed to improve the performance of the PPADs. To date, no study reported the detection of both the early as well as actual gait events. Moreover, the real-time detection algorithm was designed such that neither any event was missed nor wrongly detected that leads to 100% detection accuracy. Furthermore, the proposed algorithm was tested for both the level-ground as well as ramp walking. Except [21], other studies that dealt with both stance as well as inner-stance phases were carried out for level walking only, whereas in [21], the method was tested for ramp up and ramp down.

In this study, nine gait events were considered to be detected by the proposed algorithm. Among these, five were considered as actual gait events, namely HS, mid stance (MS), HO, TO and MSw. The remaining four were considered as early gait events, which are as follows: early HS (EHS), early MS (EMS), early HO (EHO) and early TO (ETO). The performance of the proposed method was tested for three locomotion modes (LMs), such as level ground walking (LGW), up ramp walking (URW) and down ramp walking (DRW). The detection was based on the gyroscope and accelerometer data collected from two IMUs attached to the foot and shank of the subject. A rule-based algorithm was developed to detect nine events utilizing the IMU data. The aims of this study were to design a low-cost gait phase detection unit that could work in real-time, for actual and early detection of gait events using a set of rules, and analyse the efficiency of the proposed methodology during LGW, URW and DRW.

II. Methodology

A. Experimental setup and data collection

The experimental setup consists of two IMUs, three FSRs and one microcontroller. The IMUs were attached to the shank and foot of the subject, as shown in Fig. 1(a). The gait events evaluated using FSRs were considered as the reference, and it was used to validate the performance of the proposed method. Three FSRs were attached to the bottom of the foot at three pressure points, as displayed in Fig. 1(b). The FSRs were attached to the socks using an adhesive tape, and the subjects

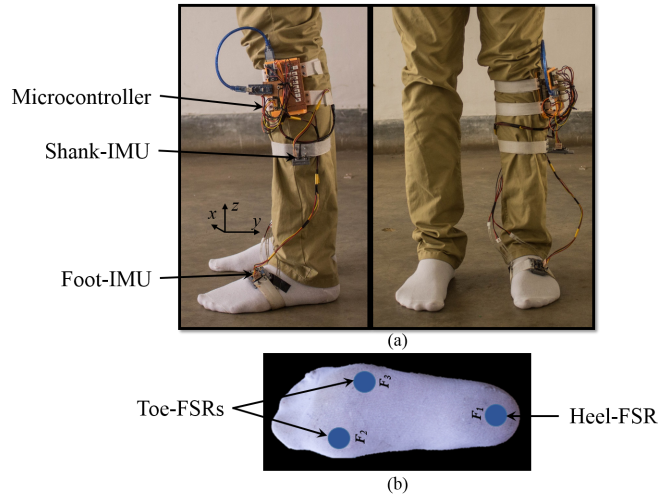


Fig. 1: (a) The complete experimental setup used for gait event detection showing the placement of IMUs and microcontroller, and (b) placement of the FSRs are shown separately. F_2 is placed at 1st Metatarsal and F_3 is placed at $4-5^{th}$ Metatarsal.

were asked to walk wearing the socks only, so that the sensors did not get detached, while walking. The FSRs were connected to the microcontroller using an ADC, and the IMUs were connected directly to the microcontroller through its I2C ports. Two 9-DOF IMUs (BNO055) and three piezoresistive force sensors (FlexiForce) with a range of 0 to 100 lbs, were chosen for the experiment. A voltage divider circuit was used to convert the FSRs output in the form of voltage, and hence, a 33 Kohm resistor was used in series for this purpose. The IMUs at the shank and foot were kept in a particular orientation, where it could yield the most accurate and stable output. Therefore, a 3-D printed housing was designed to keep the IMUs in the desired orientation. The same can be observed in Fig. 1(a).

Ten able-bodied male subjects (age: 20 - 31 years, height: 163 - 183 cm, weight: 60 - 80 kg) participated in the experiment. The subjects were asked to walk on the level ground, ramp up and ramp down terrains, while the complete setup along with FSRs was attached to the subjects left leg. The ramp considered in this study had an inclination of 12 degrees approximately. The subjects were asked to walk with three self-selected walking speeds (i.e., slow, normal and fast). Three trials for each of three walking speeds were considered for the evaluation. The total number of steps considered for analysis were 2,293, 873 and 948 for level ground, ramp up and ramp down, respectively.

B. Gait events of interest and prosthesis control

This study aims to improve the performance of lower limb PPADs. Hence, the gait events were chosen accordingly. As mentioned earlier, the gait events chosen for this study were HS, MS, HO, TO and MSw along with EHS, EMS, EHO and ETO. For human locomotion, HS defines the beginning of the stance phase. From this stage, the stiffness of joints starts increasing to support the body weight while moving forward. The joint stiffness reaches to a maximum value at

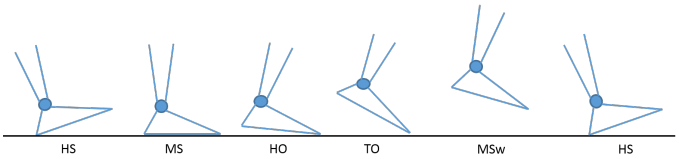


Fig. 2: The gait events chosen for this study are explained schematically. (HS: heel strike, MS: mid stance, HO: heel off, TO: toe off, MSw: mid swing, HS: heel strike)

the MS. In this study, MS was considered as the event, when the FSR attached to the heel reaches to its maximum value. The majority of the studies focused on the detection of foot flat. However, detection of MS seems to be more important for controlling PPADs, as an inadequate joint stiffness during this event might lead to loss of the balance. From HO, the stiffness of the joints starts decreasing, as the double support phase is initiated. TO indicates the end of the stance phase and also the beginning of the swing phase. MSw is considered as the phase that exists in between the TO and HS of the next step. In this phase, the ankle joint returns to its normal position and the other two joints are adjusted for ground clearance. Fig. 2 shows the schematic view of all the gait events discussed above.

The early detection of gait events (i.e., EHS, EMS, EHO and ETO) could be utilized as a prior knowledge to the state transition model. For example, the detection of EHS could be utilized to configure the torque controller in advance, so that the appropriate damping could be provided to tackle the impact caused during HS. EMS, EHO and ETO could be utilized in a similar fashion. In this way, the delay caused by the late detection, could be minimized and a better stability could be achieved using the early detection. The process flow diagram to control PPADs according to the proposed method is shown in Fig. 3. According to the flow diagram, an early detection could help the torque controller to evaluate the joint and motor kinetics and pre-activate the mechanism, while the detection algorithm recognizes the actual gait event. Similarly, the finite state controller could use the early gait events to improve the sluggish behaviour of the PPADs.

C. Rule base design

The rules used for the real-time detection of nine gait events are mentioned in Table I. Among the various outputs of two IMUs attached to the shank and foot, four had been selected for the detection of the mentioned gait events. These are shank angular velocity (SAV), foot angular velocity (FAV),

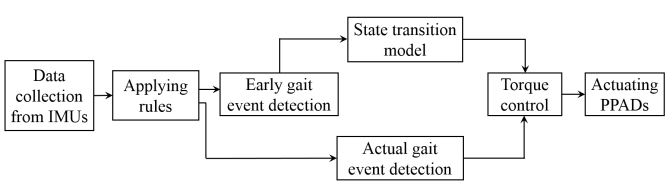


Fig. 3: The process flow diagram for controlling PPADs with the help of early detection.

TABLE I: Rules for detection of nine gait events. The variable S_{HS} is used to store the value of SAN at HS. n and (n-1) denote current and previous gait cycles, respectively.

Rule base	**Remarks**
if SAV $<-50deg/sec$ ***and*** SAV is increasing	It marks EHS
if EHS detected ***and*** SAV $>-100deg/sec$ ***and*** FAV >0 deg/sec ***and*** abs(SACX) $>1m/sec^2$	It marks HS; S_{HS} = SAN
if HS detected ***and*** FAV $<50deg/sec$ ***and*** FAV is decreasing	It marks EMS
if EMS detected ***and*** abs(FAV) $<10deg/sec$ for 3 consecutive samples	It marks MS
if MS detected ***and*** SAV is increasing	It marks EHO
if EHO detected ***and*** FAV $>1deg/sec$ for 2 consecutive samples ***and*** SAV is increasing ***and*** SAN $<0deg$ ***or*** SAN $<S_{HS}\ -25deg$	It marks HO
if HO detected ***and*** FAV $>50deg/sec$ ***and*** FAV is increasing	It marks ETO
if ETO detected ***and*** FAV $>200deg/sec$ ***or*** SAN $<S_{HS}-50deg$ ***or*** FAV $>10deg/sec$ ***and*** FAV is decreasing ***and*** SAV is increasing	It marks TO
if TO detected ***and*** SAV $<0deg/sec$	It marks MSw

shank acceleration along the x-axis (SACX) and shank angle (SAN). However, SAV and FAV were seen to have the major contributions compared to others. A calibration process was carried out before evaluating the gait events. In this process, the outputs of IMUs were set to zero, while the subjects were in standing position. The calibration process helped to eliminate the effect of gravity and removed any offset or bias from the sensors output that might be incorporated due to the misalignment of IMUs, while attaching them to the subjects leg. A duration of 20 ms was allotted for calibration at the beginning of the experiment, after which the subjects were asked to walk. The outputs of IMUs were used in their raw form to evaluate the gait events, as filtering increases the detection delay. Two or more rules were applied for the detection of an event. All such rules are discussed in Table I. The threshold values used for designing the rules in Table I were chosen empirically. A few previous studies reported less detection accuracy [21] and large delay [19] for the detection of HO compared to the other events. Therefore, special care had been taken for the detection of HO by increasing the number of rules. As a consequence, we could achieve 100% detection accuracy and a considerably less time delay for the same. In Fig. 4, three outputs of IMUs are plotted for two consecutive gait cycles, and the detected gait events based on the rules discussed in Table I, are also highlighted on the same.

D. Real-time detection and analysis

The detection algorithm collected the raw data from the IMUs, and the mentioned gait events were detected in real-time using the set of rules discussed in Table I. The outputs of heel (F_1) and toe (F_2) FSRs had been used as the reference for validation of the results obtained using the proposed algorithm.

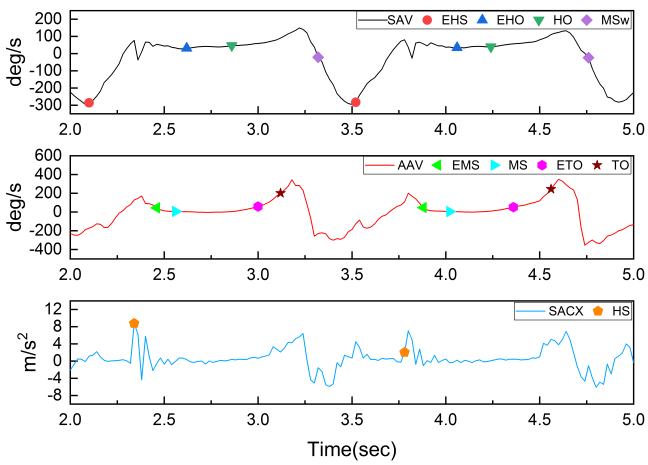


Fig. 4: Detected gait events for two consecutive gait cycles along with the variations of SAV, AAV and SACX. (SAV: shank angular velocity, AAV: ankle angular velocity, SACX: shank angular acceleration along x-axis, HS: heel strike, MS: mid stance, HO: heel off, TO: toe off, MSw: mid swing, EHS: early heel strike, EMS: early mid stance, EHO: early heel off, ETO: early toe off).

A separate set of rules was configured for extracting the gait events from the FSR output. Using FSR, HS was detected, when the voltage across $F_1 > Th_1$, and then the algorithm checked the same condition for three consecutive samples to confirm that the detected sample was actually an HS. MS was detected, when F_1 reached its maximum value. HO was detected after MS, when the value of F_1 reduced to zero. Similarly, TO was detected after HO, when $F_2 < Th_1$. The value of Th_1 was kept equal to 1 V. MSw was evaluated as follows: $MSw = (HS(n) - TO(n-1))/2$, where n and $(n-1)$ represent the current and the previous gait cycles, respectively. Alike IMU, no filtering was used for FSR output. The use of smoothening or low pass filter for FSR output was seen to increase the delay in the detection.

The real-time algorithm could evaluate the gait events from both the IMUs as well as FSRs. After getting the two results, the performance of the proposed method was evaluated in terms of detection accuracy, time delay and early detection time. The detection accuracy and time delay were evaluated between the output of the proposed algorithm and that of the FSRs, considering the outputs of FSRs as the standard. The time delay (T_d) and early detection time (T_e) for each event were calculated as follows: $T_d = t_{act} - t_{FSR}$ and $T_e = t_{early} - t_{FSR}$, respectively, where t_{act}, t_{early} and t_{FSR} are the detected actual gait event using the proposed approach, early event and gait event detected using FSR, respectively. The detection accuracy was calculated after evaluating the number of detected events from the proposed and reference approaches.

III. Results

Two parameters had been used in the literature to evaluate the performance of a gait event detection method, namely detection accuracy and time delay. The obtained detection accuracy using the proposed rule-based algorithm was found to be equal to 100 % for all the gait events. The time difference between the proposed method and the reference one was analyzed using different statistical techniques discussed below.

A. Analysis of time delay and early detection time

The time delay and early detection time were evaluated, as discussed in Section II(D). The mean delay and standard deviation (SD) of nine events considering all the subjects were evaluated separately for LGW, URW and DRW, as shown in Table II. In Table II, the overall mean and SD for three LMs are also provided. The positive and negative mean values indicate the delayed and early detection, respectively. By observing the results in Table II, it was found that among the actual events, HS, MS and HO were detected late, and TO and MSw were detected early, whereas the mean was seen to be negative for all the early events.

During the stance phase, HS and TO exhibited the smaller mean delays compared to MS and HO events for all three LMs. However, the standard deviation for HS was seen to be smaller than TO. The overall mean delay for the detection of HS evaluated across three LMs was found to be the minimum among the actual gait events. The same was found to be maximum for HO followed by MS, TO and HS. The mean delay for HS during URW and TO during DRW was seen to be the minimum among three LMs. Similarly, the same for MS during LGW and HO during URW was found to be the minimum. Among the three LMs, the proposed algorithm performed the best during LGW in terms of the mean and standard deviation of delay. A variation in the mean delay for a few events across three LMs was also observed. For example, mean time delay for MS during LGW, URW and DRW were 20.6 ms,42.8 ms and 92.8 ms, respectively. The same was also true for HO, whereas the variations were found to be less for HS, TO and MSw.

The mean delay and standard deviation were found to be less for HS and TO, as they represent the starting and terminal stance phases, respectively. Being inner-stance phases, the delay for MS and HO was seen to be greater than HS and TO. A large distribution of time delay was observed for a few events due to the consideration of a variety of self-selected walking speeds (slow, normal and fast) among ten subjects during the experiment. It had been observed that a slow walking speed generates a large delay, whereas fast walking speed reduces the delay and sometimes detects the events early. The patterns of the gait cycle for walking on three different LMs mentioned in this study are not identical. This might be the reason for the variations of results across different LMs.

For the early detection, the average early detection time evaluated considering all three LMs was found to be the maximum for EHS, and minimum for EMS. A high early detection time would give more time to the state transition model to respond to change in gait phase during walking using PPADs, whereas a small early detection time might demand high computation speed for the same. An early gait event

TABLE II: Mean ± 1 SD (ms) values of time difference between the proposed and reference methods for all nine events across three LMs.

LM	HS (ms)	MS (ms)	HO (ms)	TO (ms)	MSw (ms)	EHS (ms)	EMS (ms)	EHO (ms)	ETO (ms)
LGW	10.4 ± 26.5	20.6 ± 86	49.1 ± 85	-13.7 ± 76.6	-97.6 ± 37	-165.8 ± 66	-95.9 ± 52	-169.5 ± 108	-116.8 ± 46
URW	7.6 ± 35.4	42.8 ± 76	32.3 ± 110	-22.8 ± 80	-78.3 ± 78	-168 ± 53	-86.2 ± 66	-131 ± 156	-125 ± 77
DRW	16.9 ± 25	92.8 ± 86	87.2 ± 105	-5.5 ± 83	-82.4 ± 75	-162 ± 53	-37 ± 67	-58.9 ± 174	84.5 ± 75
All LMs	11.8 ± 28	41.7 ± 82	51.7 ± 87	-16.4 ± 63	-90 ± 59	-165.5 ± 61	-80.3 ± 63	-135.9 ± 144	-111.2 ± 67

was detected before the respective actual event and after the previous actual event. For example, EHO was detected before HO and after MS. Hence, it is not possible to detect a few gait events very early due to the fact that the time gap between two consecutive actual gait events was small. This phenomenon led to the low early detection time for EMS, as the time gap between HS and MS was less. Alike HS and HO, EHS and EHO were seen to possess the minimum and maximum variations of early detection time, respectively. However, the variations were seen to be similar for EHS, EMS and ETO.

The detected reference, actual and early gait events are shown in Fig. 5 along with the variations of FSRs for five consecutive gait cycles during LGW. From Fig. 5, it can be seen that the actual gait events detected from the reference method and the proposed one were close to one another, whereas the early detection always occurred before the respective actual one and after the previous actual gait event.

B. Distribution of time delay and early detection time

In order to observe the distribution of time difference, boxplots were evaluated for nine gait events across three LMs, and the same is shown in Fig. 6. The interquartile regions were found to be close to the zero reference line for all actual gait events. Similarly, the major distribution of time delay for all early detections belongs to the negative side of the zero reference line and away from it. The above phenomena were true for all three LMs. These observations conclude that the actual events were detected with the minimum time difference, and for the early detection, the difference between the actual and early events was large. Thus, the early detection time could be utilized for improving the response of PPADs. Moreover, either almost no or very less overlapping was observed between the interquartile ranges of an actual event and its early detection, which concludes that an early detection generally occurred before the actual one. Looking at the plots, it can also be seen that HS and TO showed the less variability (i.e., stretch between minima and maxima of the boxplot) among actual gait events for all three LMs. Similarly, ETO had shown the lowest variability among the early gait events, whereas EHO was seen to exhibit the highest variations of delay. A few outliers could be observed for all the events, whereas the maximum outliers were seen for MSw, as the reference (using FSRs) of this event was measured indirectly. However, looking at the error distributions and outliers in Fig. 6, it can be concluded that the proposed algorithm worked better for LGW and URW compared to DRW.

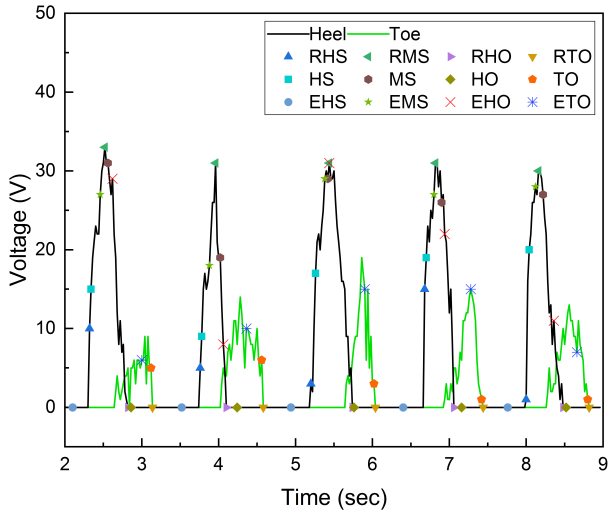
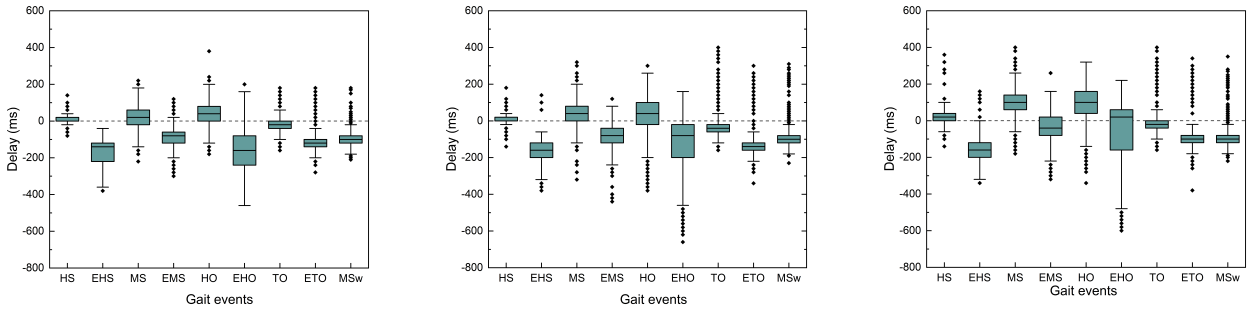


Fig. 5: The actual and early gait events evaluated for five consecutive gait cycles during LGW and plotted along the variations of F_1 and F_2. The force exerted on the sensors is represented along y-axis in form of output voltage. RHS, RMS, RHO and RTO denote the reference events detected from the FSRs.

C. Analysis of agreement between the reference and proposed methods

The Bland-Altman plot shows the agreement between the proposed and the reference methods. The Bland-Altman plots were evaluated for actual events (i.e., HS, MS, HO and TO) during LGW, URW and DRW and shown in Figs.7(a), (b) and (c), respectively. As the reference for early detections was not available, the Bland-Altman plot was not evaluated for the same. In this plot, the time differences for gait event detection evaluated using the two methods discussed above were plotted against their averages. The Bland-Altman plots also showed the mean and 95% confidence intervals of time differences. In Fig. 7, it can be seen that the mean time delay for HS and TO was closer to zero compared to MS and HO for three LMs. However, the 95% confidence intervals for TO were seen to be the higher than that of HS. Across three LMs, the overall performances for LGW and URW were seen to be the better compared to DRW in terms of the mean difference. However, in terms of outliers, the LGW was found to be the best. In spite of having 2,293 data points, the number of outliers was seen to be less for the Bland-Altman plot during LGW compared to other two LMs. For HS and TO, the variations in mean and 95% confidence intervals across different LMs were found to



(a) (b) (c)

Fig. 6: The boxplots for nine gait events across (a) LGW, (b) URW and (c) DRW. The dotted line in the graphs indicates the zero reference line. The line inside the box denotes the median of error.

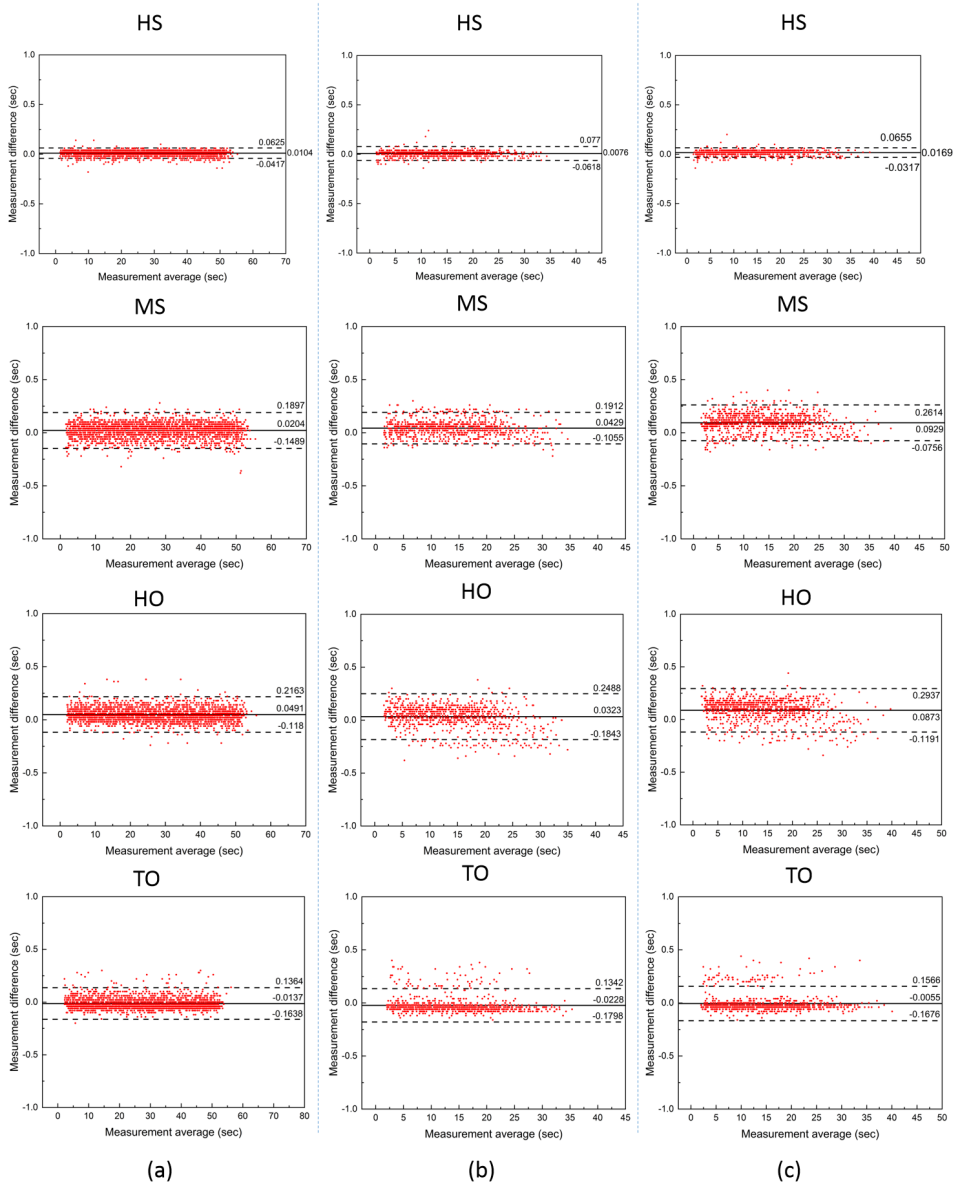


Fig. 7: The Bland-Altman plots for actual gait events across (a) LGW, (b) URW and (c) DRW. The solid lines indicate the mean difference and the dashed lines denote the 95% confidence intervals i.e., mean $\pm$ 1.96 SD.

be less, whereas the same was found to be more in the case of MS and HO events.

IV. Discussion

The performance of the proposed method was examined using three statistical approaches discussed above. The conclusion drawn from them suggested that the starting (HS) and terminal (TO) stance phases performed well compared to inner stance phases, i.e., MS and HO. The higher mean value and larger distribution of time delay were seen for the inner stance phases compared to HS and TO. The independent selection of walking speeds, i.e., slow, normal and fast among 10 subjects might be one of the reasons for the large distribution of time delay. Moreover, the noise due to the fluctuations in gyroscope and accelerometer readings, FSR circuitry and slip of IMUs and FSRs from their desired position might also cause the greater distribution of time delay. The results also concluded that the algorithm worked better for LGW compared to ramp walking, and a noticeable difference was found for DRW. This could be due to the unfamiliarity of the subjects to walk on the ramp. Normal subjects spend most of the time walking on the level ground, whereas the ramp is encountered rarely. This might cause dissimilar gait patterns among the subjects, and hence, variations in the accelerometer and gyroscope readings were observed. Moreover, the gait patterns for ramp and level ground walking were not exactly identical. One may consider more training and separate threshold values for the rules to improve performance during ramp walking.

In this study, the total number of events detected using FSRs was found to be equal to 20,570 considering HS, MS, HO, TO and MSw, and the number of events detected by the proposed algorithm was found to be the same. Hence, the detection accuracy was obtained as 100% for all the events. Moreover, early events were always detected before the respective actual events. Only a few studies that dealt with the detection of stance and inner-stance phases reported the accuracy in terms of detection of gait events. Maqbool et al. [21] reported 100% detection accuracy for all four events except HO. However, in this study, the rule-base was designed in such a way that it detected HOs correctly, and no incident of misdetection was found. Similarly, in [22], the detection accuracy was reported as 99.2% for the detection of nine gait events that include the inner-stance and inner-swing phases. Among the studies that dealt with HS and TO only, Gouwanda et al. [13] and Maqbool et al. [11] reported 100% detection accuracy. However, Gouwanda et al. [13] did not consider ramp walking. Similarly, Catalfamo [14] reported the detection accuracy as 99.5% for the level-ground and ramp walking. In [23], the detection accuracy was reported as 99.6% for LGW only.

In this study, the mean time delay for stance and inner stance phases with respect to the reference system was found to be as 10.4 $\pm$ 26.5 ms, 20.6 $\pm$ 86 ms, 49.1 $\pm$ 85.3 ms, -13.7 $\pm$ 76.6 ms for HS, MS, HO and TO, respectively, during LGW. Among the other studies that dealt with both the stance as well as inner stance phases during LGW, Maqbool et al. [21] reported the time delay as 16 $\pm$ 9 ms, -3.6 $\pm$ 49 ms and -16 $\pm$ 15.9 ms, for HS, HO and TO, respectively. Similarly, Lambrecht et al. [19] reported the same as 11.10 $\pm$ 10.72 ms, 189.39 $\pm$ 44.99 ms and 6.40 $\pm$ 15.49 ms for HS, HO and TO using RTGSD-G6 algorithm, which was one out of the three algorithms they proposed. Similarly, by using RTGSD-B6 algorithm, the time delay for the same gait events was reported as 26.53 $\pm$ 22.43 ms, 3.63 $\pm$ 110.37 ms and 38.52 $\pm$ 25.35 ms for HS, HO and TO, respectively [19]. Another study carried out by Mariani et al. [18] reported the time delay as 1 $\pm$ 13 ms, 4 $\pm$ 54 ms, and -3 $\pm$ 13 ms for HS, HO and TO, respectively. Bouttayamou et al. [17] reported the delay as 1.3 $\pm$ 7.2 ms, -3.7 $\pm$ 14.5 ms and -1.8 $\pm$ 11.8 ms, for HS, HO and TO, respectively. The time delay reported in [17] and [18] had good accuracy and precision, however, their approaches were implemented offline. Pappas et al. [3] reported the mean time delay as 70 ms, 40 ms and 35 ms for HS, HO and TO, respectively. Chia et al. [10] reported the same for HS and TO as 13 and 10 ms, respectively. The study in [11], reported the delay of 10.7 $\pm$ 17.9 ms and -7.6 $\pm$ 35.2 ms for HS and TO, respectively. In this study, we considered MS instead of detecting foot flat and the time delay for the same had not been reported in any other studies. It has been noticed that for some events, the proposed study showed the better results in terms of mean delay but for some others, it did not, while compared to the studies that dealt with real-time detection of the gait events. For example, the absolute mean delay in the proposed study was found to be less than that reported in [21] for HS and TO, [19] for HS and HO using RTGSD-G6 algorithm, [19] for HS and TO using RTGSD-B6 algorithm, [3] for HS and TO, [10] for HS and [11] for HS. Similarly, it was found to be higher than that reported in [21] for HO, [19] for TO using RTGSD-G6 algorithm, [19] for HO using RTGSD-B6 algorithm, [3] for HO, [10] for TO and [11] for TO. However, the number of samples used to evaluate the proposed study during LGW (i.e., 2,293 samples per event) was higher than that of the other studies mentioned above. The standard deviation was found to be high due to the variations of walking speeds for 10 subjects, where each of them was walking with three self-selected speeds (slow, normal and fast).

In the proposed study, the time delay for ramp walking was found to be as 7.6 $\pm$ 35.4 ms, 42.8 $\pm$ 75.7 ms, 32.3 $\pm$ 110 ms and -22.8 $\pm$ 80 ms for HS, MS, HO and TO, respectively, during URW; and 16.9 $\pm$ 24.8 ms, 92.8 $\pm$ 85.9 ms, 87.2 $\pm$ 105.3 ms and -5.5 $\pm$ 82.7 ms for HS, MS, HO and TO, respectively, for DRW. To the best of authors knowledge, only Maqbool et al. [21] investigated the stance and inner stance phases for ramp walking. They reported the time delay for HS, HO and TO as 18.8 $\pm$ 11.6 ms, -42.8 $\pm$ 57 ms and -17.2 $\pm$ 21.3 ms, respectively, for URW, and for DRW, the same was reported as 17 $\pm$ 11 ms, 5.7 $\pm$ 52.6 ms and -22.7 $\pm$ 19.4 ms, respectively. Compared to [21], the proposed study performed better in terms of mean delay except for the events TO during URW and HO during DRW. Catalfamo et al. [14] investigated the starting and terminal stance phases for ramp walking. They reported the time delay for HS and TO as -21 $\pm$ 15 ms and 43 $\pm$ 10 ms, respectively, for URW, and for the DRW, they obtained -9 $\pm$ 20 ms and 73 $\pm$ 12 ms time delay for HS and TO, respectively. Compared to [14], the proposed

study performed better for HS and TO events during URW. However, for DRW, the proposed study showed the smaller mean absolute delay for the detection of TO, but for HS, the mean absolute delay was found to be less in [14]. Similarly, in [11], the time delay for URW was reported as 14 $\pm$ 21 ms and -5 $\pm$ 32 ms for HS and TO, respectively; and for DRW, it was reported as 10.5 $\pm$ 17 ms and -25 $\pm$ 36 ms for HS and TO, respectively. In the proposed study, the time delay was seen to be less for HS during URW and TO during DRW compared to the delay reported in [11]. However, for the other two events, i.e., HS during DRW and TO during URW, the mean delay was found to be less in [11]. In case of ramp walking, the SD was found to be high for a few events for the same reason as mentioned during LGW.

The average early detection time during LGW, URW and DRW was found to be equal to 123 ms. As discussed in Section II(B), this early detection time can be utilized to improve the real-time performance of PPADs. In this way, the proposed approach might help to achieve the better stability, while using a PPAD. Moreover, for fast walking using PPADs, this approach might be advantageous. The only disadvantage of this study was the usage of two IMUs. However, for the detection of nine gait events, only one IMU could have been insufficient, or it might affect the results. Hence, the authors decided to use two IMUs. The limitation of this study lies with the fact that the standard deviation of time delay was more than 100 ms for HO during URW and DRW, and the mean delay for MS (92.8 ms) and HO (87.2 ms) during DRW was found to be higher compared to the other events.

V. Conclusion

In this study, an early detection method for gait event detection was proposed and tested with ten control subjects. The utilization of the early detection for improving the performance of PPADs had also been discussed. A low-cost standalone prototype was designed to detect a total of nine gait events in real-time. Moreover, the simplicity had also been maintained by using a set of rules for the detection of gait events. The proposed rule-based algorithm was tested for three different LMs, such as LGW, URW and DRW. The results showed that the detection accuracy was 100% for all gait events. The time delay was evaluated with respect to a standard method that uses FSRs. It was found to be similar and even better in some cases compared to that of the state-of-the-art methods. For early gait events, the early detection time was evaluated instead of time delay, which could be utilized to improve the response time of PPADs. In future, the authors will be working on the detection of actual and early gait phases for stair ascend and descend. Moreover, the effect of early detection on controller settling time will be investigated with the help of some practical experiments.

References

[1] D. Zlatnik, B. Steiner, and G. Schweitzer, "Finite-state control of a trans-femoral (tf) prosthesis," *IEEE Transactions on Control Systems Technology*, vol. 10, no. 3, pp. 408–420, 2002.

[2] S. K. Au, J. Weber, and H. Herr, "Powered ankle–foot prosthesis improves walking metabolic economy," *IEEE Transactions on Robotics*, vol. 25, no. 1, pp. 51–66, 2009.

[3] I. P. Pappas, M. R. Popovic, T. Keller, V. Dietz, and M. Morari, "A reliable gait phase detection system," *IEEE Transactions on neural systems and rehabilitation engineering*, vol. 9, no. 2, pp. 113–125, 2001.

[4] J. A. Blaya and H. Herr, "Adaptive control of a variable-impedance ankle-foot orthosis to assist drop-foot gait," *IEEE Transactions on neural systems and rehabilitation engineering*, vol. 12, no. 1, pp. 24–31, 2004.

[5] P. Cherelle, K. Junius, V. Grosu, H. Cuypers, B. Vanderborght, and D. Lefeber, "The amp-foot 2.1: actuator design, control and experiments with an amputee," *Robotica*, vol. 32, no. 8, pp. 1347–1361, 2014.

[6] A. T. M. Willemsen, F. Bloemhof, and H. B. Boom, "Automatic stance-swing phase detection from accelerometer data for peroneal nerve stimulation," *IEEE Transactions on biomedical engineering*, vol. 37, no. 12, pp. 1201–1208, 1990.

[7] G. M. Lyons, T. Sinkjær, J. H. Burridge, and D. J. Wilcox, "A review of portable fes-based neural orthoses for the correction of drop foot," *IEEE Transactions on neural systems and rehabilitation engineering*, vol. 10, no. 4, pp. 260–279, 2002.

[8] R. Williamson and B. J. Andrews, "Gait event detection for fes using accelerometers and supervised machine learning," *IEEE Transactions on Rehabilitation Engineering*, vol. 8, no. 3, pp. 312–319, 2000.

[9] Y. C. Han, K. I. Wong, and I. Murray, "Gait phase detection for normal and abnormal gaits using imu," *IEEE Sensors Journal*, vol. 19, no. 9, pp. 3439–3448, 2019.

[10] N. C. Bejarano, E. Ambrosini, A. Pedrocchi, G. Ferrigno, M. Monticone, and S. Ferrante, "A novel adaptive, real-time algorithm to detect gait events from wearable sensors," *IEEE transactions on neural systems and rehabilitation engineering*, vol. 23, no. 3, pp. 413–422, 2014.

[11] H. F. Maqbool, M. A. B. Husman, M. I. Awad, A. Abouhossein, N. Iqbal, and A. A. Dehghani-Sanij, "A real-time gait event detection for lower limb prosthesis control and evaluation," *IEEE transactions on neural systems and rehabilitation engineering*, vol. 25, no. 9, pp. 1500–1509, 2016.

[12] H. Zhou, N. Ji, O. Samuel, Y. Cao, Z. Zhao, S. Chen, and G. Li, "Towards real-time detection of gait events on different terrains using time-frequency analysis and peak heuristics algorithm," *Sensors*, vol. 16, no. 10, p. 1634, 2016.

[13] D. Gouwanda and A. A. Gopalai, "A robust real-time gait event detection using wireless gyroscope and its application on normal and altered gaits," *Medical engineering & physics*, vol. 37, no. 2, pp. 219–225, 2015.

[14] P. Catalfamo, S. Ghoussayni, and D. Ewins, "Gait event detection on level ground and incline walking using a rate gyroscope," *Sensors*, vol. 10, no. 6, pp. 5683–5702, 2010.

[15] J. C. Pérez-Ibarra, A. A. Siqueira, and H. I. Krebs, "Real-time identification of gait events in impaired subjects using a single-imu foot-mounted device," *IEEE Sensors Journal*, 2019.

[16] A. Mannini, V. Genovese, and A. M. Sabatini, "Online decoding of hidden markov models for gait event detection using foot-mounted gyroscopes," *IEEE journal of biomedical and health informatics*, vol. 18, no. 4, pp. 1122–1130, 2013.

[17] M. Boutaayamou, C. Schwartz, J. Stamatakis, V. Denoël, D. Maquet, B. Forthomme, J.-L. Croisier, B. Macq, J. G. Verly, G. Garraux *et al.*, "Development and validation of an accelerometer-based method for quantifying gait events," *Medical engineering & physics*, vol. 37, no. 2, pp. 226–232, 2015.

[18] B. Mariani, H. Rouhani, X. Crevoisier, and K. Aminian, "Quantitative estimation of foot-flat and stance phase of gait using foot-worn inertial sensors," *Gait & posture*, vol. 37, no. 2, pp. 229–234, 2013.

[19] S. Lambrecht, A. Harutyunyan, K. Tanghe, M. Afschrift, J. De Schutter, and I. Jonkers, "Real-time gait event detection based on kinematic data coupled to a biomechanical model," *Sensors*, vol. 17, no. 4, p. 671, 2017.

[20] P. Muller, T. Steel, and T. Schauer, "Experimental evaluation of a novel inertial sensor based realtime gait phase detection algorithm," in *Proceedings of the Technically Assisted Rehabilitation Conference*, 2015.

[21] H. F. Maqbool, M. A. B. Husman, M. I. Awad, A. Abouhossein, N. Iqbal, M. Tahir, and A. A. Dehghani-Sanij, "Heuristic real-time detection of temporal gait events for lower limb amputees," *IEEE Sensors Journal*, vol. 19, no. 8, pp. 3138–3148, 2018.

[22] U. Martinez-Hernandez, I. Mahmood, and A. A. Dehghani-Sanij, "Simultaneous bayesian recognition of locomotion and gait phases with wearable sensors," *IEEE Sensors Journal*, vol. 18, no. 3, pp. 1282–1290, 2017.

[23] A. Salarian, H. Russmann, F. J. Vingerhoets, C. Dehollain, Y. Blanc, P. R. Burkhard, and K. Aminian, "Gait assessment in parkinson's disease: toward an ambulatory system for long-term monitoring," *IEEE transactions on biomedical engineering*, vol. 51, no. 8, pp. 1434–1443, 2004.